

Dis. Dig. Mapa de Karnaugh 2

1) $F(ABC) = 1$ se $ABC \geq 5$

A	B	C	F	A \ BC
0	0	0	0	00
0	0	1	0	01
0	1	0	0	10
0	1	1	0	11
1	0	0	0	00
1	0	1	1	01
1	1	0	1	10
1	1	1	1	11

Simplificando: $AB + AC = A(B + C)$

2) $F(A, B, C) = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C$

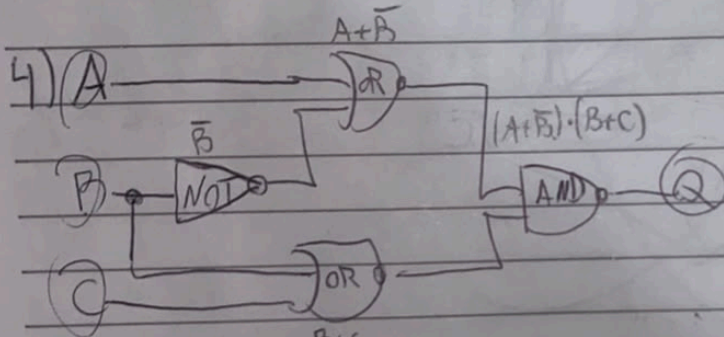
A \ BC	00	01	11	10
0		1	1	
1	1	1		

Simplificando: $A\bar{B} + \bar{A}C$

3)

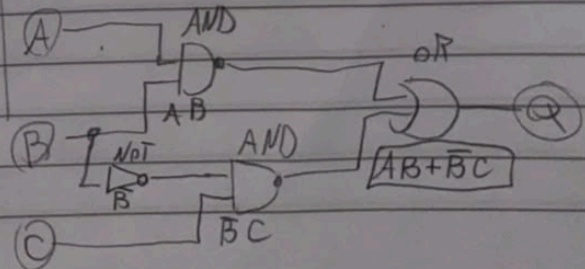
AB \ CD	00	01	11	10
00	1			1
01				
11	1	1	1	1
10	1			1

Simplification: $AB + \bar{B}\bar{D}$



A	B	C	$A \vee B$	$B \vee C$	$x \wedge y$	A \ BC
0	0	0	1	0	0	00
0	0	1	1	1	1	01
0	1	0	0	1	0	11
0	1	1	0	1	0	10
1	0	0	1	0	0	
1	0	1	1	1	1	
1	1	0	1	1	1	
1	1	1	1	1	1	

Simplification: $AB + \bar{B}C$



5) Tabela V - Mintermos

a) SOP (mintermos)

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$\overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C}$$

b) Mapa

		BC			
A		00	01	11	10
		0	1		1
1		1	1		1

$$\overline{A}\overline{B}$$

$$\overline{C}$$

c) Simplificado: $\overline{A}\overline{B} + \overline{C}$

1ª Expressão: possui 11 portas

2ª Expressão Simplificada: possui 4 portas

6) $\overline{A}BCD + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} + A\overline{B}C\overline{D} + A\overline{B}C\overline{D} + A\overline{B}C\overline{D} + A\overline{B}C\overline{D}$

0 0010 0101 0111 1000 1010 1101 1

AB \ CD	00	01	11	10
00	1			1
01		1	1	
11		1	1	
10				1

Simplificando: $\overline{B}\overline{D} + \overline{B}D$

7) Maxtermas PoP

$F(A,B,C) = (A+B+C)(A+B+\overline{C})(\overline{A}+B+C)(\overline{A}+\overline{B}+C)$

0 0 0 0 0 1 1 0 0 1 1 0

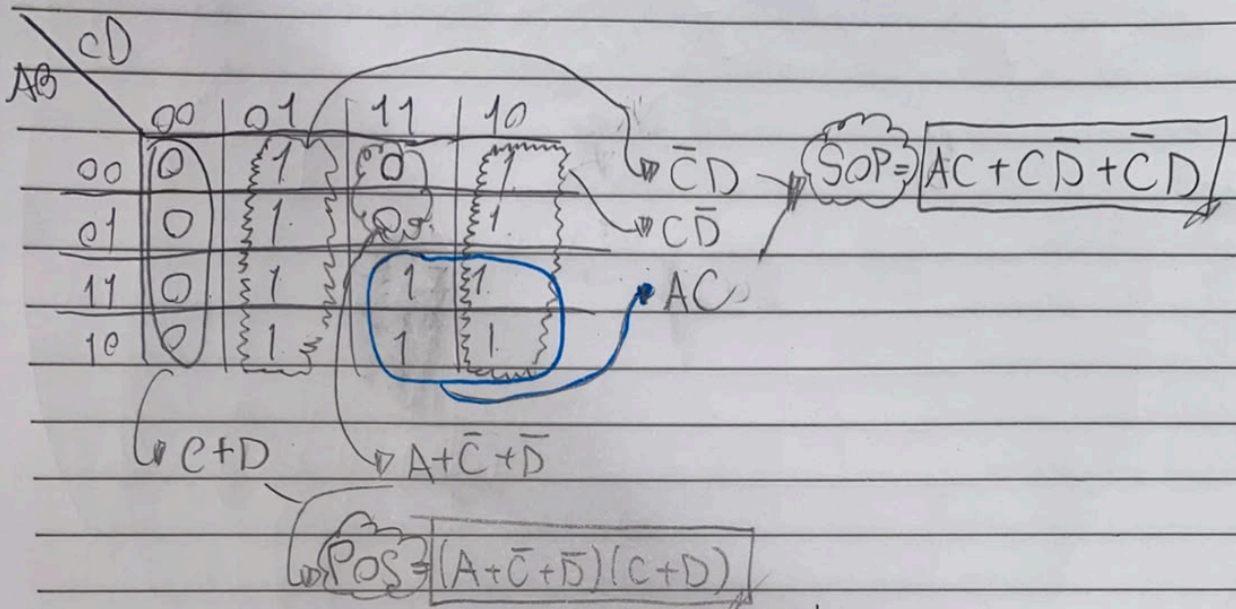
A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

A \ BC	00	01	11	10
0	0	0	1	1
1	0	1	1	0

$(A+B)$ $(\overline{A}+C)$ $(B+C)$

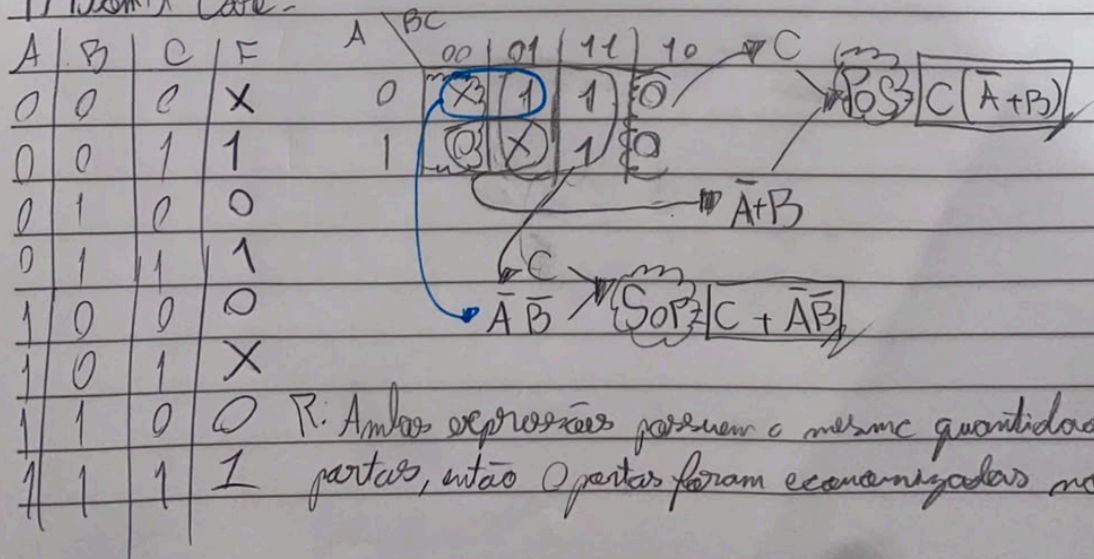
Simplificando $(A+B)(\overline{A}+C)(B+C)$

8) $(A+B+C+D)(A+B+\bar{C}+\bar{D})(A+\bar{B}+C+D)(A+\bar{B}+\bar{C}+\bar{D})(\bar{A}+B+C+D)(\bar{A}+\bar{B}+C+D)$
 0 0 0 1 1 0 1 0 0 0 1 1 1 1 0 0 0 1 1 0 0



R: A solução com com menos custo é o POS, pois possui 3 portas e 7 saídas, já SOP tem 4 portas e 9 saídas.

9) Don't Care:



R: Ambas expressões possuem a mesma quantidade de portas, então 0 portas foram economizadas nos 2 casos.

